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Paper Code : 21312

F–412

B. C. A. (Third Semester)
EXAMINATION, 2017-18
(New Course)

Paper No. BCA–302–N

COMPUTER ORGANISATION

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks.

1. (a) Simplify the following Boolean Function in product of sums form by mean of a four variable map. Draw the logic diagram with (i) OR-AND gates (ii) NOR gates $F(w, x, y, z) = \Sigma(2, 3, 4, 5, 6, 7, 11, 14, 15)$ 10

   (b) Show that a JK flip-flop can be converted to a Δ flip-flop with an inverter between the J and K inputs. 4

2. Explain Booth's algorithm for multiplication of two numbers. Show the step by step multiplication process of (+14) * (+ 13) by using Booth's algorithm. 14

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3. What do you mean by addressing modes ? Explain the following addressing modes : 4, 10
   (i) Implied mode
   (ii) Immediate mode
   (iii) Autoincrement mode
   (iv) Register indirect mode
   (v) Index Addressing mode

4. What is instruction format ? Evaluate the arithmetic statement : 14

   $$X = (A + B) * (C + D)$$

   by using :
   (i) Three address instructions
   (ii) Two address instructions
   (iii) One address instructions
   (iv) Zero address instructions
   (v) RISC instructions

5. (a) Explain memory hierarchy in a computer a system. 7

   (b) Explain the following : $3\frac{1}{2}$ each
   (i) Static and Dynamic RAMs
   (ii) PROM and EPROM

6. (a) What is hit ratio ? Explain Associative mapping or Direct mapping for cache memory. 7

   (b) What is page fault ? Explain page replacement for a virtual memory system. 7

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7. Explain the following : $3\frac{1}{2}$ each

   (i) Bus arbitration

   (ii) DMA

   (iii) Types of interrupts

   (iv) Programmed I/O

8. Write short notes on any *four* of the following : $3\frac{1}{2}$ each

   (i) Handshaking

   (ii) Processor Organization

   (iii) BCD code

   (iv) MMV

   (v) Program Counter

   (vi) Universal gates

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